Tour de Maths 2024

Leg 5

Dainfern College

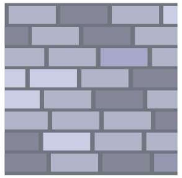
Question Paper

https://www.piximaths.co.uk/problem-solving

5 Mark Questions

Question 1

David has a rectangular paved area that he wants to enlarge. He increases the length by 30% and width by 20%. By what percentage has the total area of the paved section increased?



Question 2

The average of six numbers is 4. A seventh number is added and the new average is 5. What is the value of the seventh number that was added?

Question 3

In a certain village, there are 800 women. 3% of the women wear one earring. Of the other 97%, half of them wear two earrings and the other half wear none. How many earrings are worn altogether?

Question 4

Thando writes a sequence of numbers. The first number is 156 and each number after that is 4 less than the number before it. How many numbers must be added so that the sum of the sequence of numbers is 0?

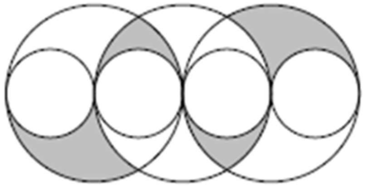
Question 5

The product of the digits of a 10-digit number is 15. What is the sum of the digits of this number?

Question 6

The diagram shows three large circles of equal radius with 4 small circles of equal radius.

The centres of all circles and all points of contact lie on one straight line. The radius of each small circle is 1.



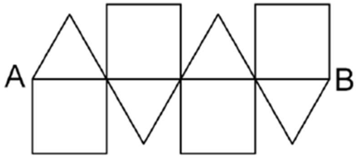
What is the value of the shaded area?

(a) π (b) $\frac{3}{2}\pi$ (c) 2π (d) 3π (e) 4π (f) 6π

Question 7

A piece of wire is used to make the shape below. The shape consists of four identical squares and four identical equilateral triangles.

If the length from A to B is 30cm, how many centimetres of wire is needed to make the shape?

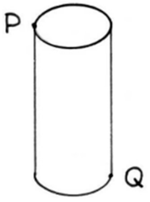


(a) 150 (b) 180 (c) 165 (d) 215 (e) 220

Question 8

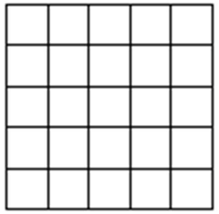
The diagram shows a circular cylinder with a circumference of 6 units and a height of 4 units. Points P and Q are diametrically opposite.

What is the shortest distance along the surface of the cylinder (in simplified form)?



Question 9

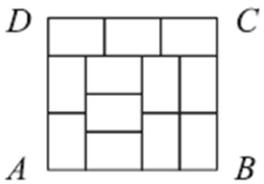
What is the smallest number of cells that need to be coloured in the 5x5 square so that any 1x4 or 4x1 rectangle lying inside the square has at least one cell coloured in?



Question 10

The diagram alongside shows a large rectangle ABCD divided into 12 identical small rectangles.

What is the ratio of AD:DC?



(a) $\frac{8}{9}$ (b) $\frac{5}{6}$ (c) $\frac{7}{8}$ (d) $\frac{2}{3}$ (e) $\frac{9}{8}$

10 Mark Questions

Question 11

$(a^{-1}+b^{-1})^{-1}$ when **fully simplified** is equal to:

(a) $a^{-2}+b^{-2}$ (b) $\frac{ab}{a+b}$ (c) $a+b$ (d) $-a-b$ (e) $\frac{a+b}{ab}$

Question 12

The operation $\otimes$ is defined for all non-zero numbers by: $a \otimes b = \frac{a^2}{b}$.

Determine the value of: $\big((1 \otimes 2) \otimes 3\big) - \big(1 \otimes (2 \otimes 3)\big)$.

Give your answer as a simplified fraction.

Question 13

The integers greater than 1 are written in a pattern as shown:

Row 1: 2

Row 2: 3 4

Row 3: 5 6 7

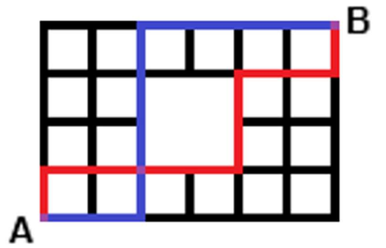
Row 4: 8 9 10 11

etc

What is the last number in the 20th row?

Question 14

How many different paths from A to B are there in the network shown, if you may only move up and to the right along the lines?



Question 15

James is competing in a triathlon.

This particular triathlon consists of a 5km swim, a 40km cycle and a 10km run.

He can swim at an average speed of 2,5km/h and cycle at an average speed of 25km/h. There are also two transition stages between each discipline (between the swim and the cycle and between the cycle and the run) that are 4 minutes each.

What speed (in km/h) must James average on the final run to finish the triathlon in 5 hours?

(leave your answer as a simplified fraction)

20 Mark Questions

Question 16

There are 16 children in a class taking part in a lucky draw. Two names are chosen at random, one after the other, without replacement. The chance that they are both girls is 55%.

How many boys are in the class?

Question 17

Jacob decided to enter numbers into the cells of a 3x3 table so that the sum of the numbers in all four possible 2x2 squares will be the same.

The numbers in three of the corner cells have already been given. What number should be written in the fourth corner cell?

2		4
?		3

(a) 0 (b) 1 (c) 4 (d) 5 (e) 6

Question 18

Given $a^{2x} \times a^{y} = a^{3}$ and $\frac{(a^{3})^{x}}{a^{4y}} = a^{32}$.

Solve for x and y and then find the product of x and y.

Question 19

Determine the **exact** value of:

$$\pi\sqrt{\pi\sqrt{\pi\sqrt{\pi\sqrt{\pi \ldots}}}}$$

Question 20

The integers 1,2,3,4,5,6,7,8 and 9 can be used to fill the boxes with a base and a power.

How many **different** solutions can be found so that no digit is used more than once in the same equation?